# Justifying constructions using the properties of quadrilaterals

**1**

**a**

**To prove: $\overleftrightarrow{AB}$ is constructed parallel to $\overleftrightarrow{DC}$**

| | Statements | Reasons |
|---|---|---|
| **1.** | $\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$ | The arcs that created the segments have the same radius. |
| **2.** | $ABCD$ is a paralellogram | If a quadrilateral has both pairs of opposite sides congruent, then it is a parallelogram. |
| **3.** | $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$ | Definition of a parallelogram. |

**b**

**To prove: $\overleftrightarrow{CD}$ is constructed so that $\overline{AB} \perp \overline{CD}$**

| | Statements | Reasons |
|---|---|---|
| **1.** | $\overline{AC} \cong \overline{CB} \cong \overline{BD} \cong \overline{DA}$ | The arcs that created the segments have the same radius. |
| **2.** | $\overline{ACBD}$ is a rhombus | Definition of a rhombus |
| **3.** | $\overline{AB} \perp \overline{CD}$ | If a quadrilateral is a rhombus, then its diagonals are perpendicular. |

**c**

**To prove: $\overleftrightarrow{CD}$ is constructed so that $\overline{AE} \cong \overline{BE}$**

| | Statements | Reasons |
|---|---|---|
| **1.** | $\overline{AC} \cong \overline{CB} \cong \overline{BD} \cong \overline{DA}$ | The arcs that created the segments have the same radius. |
| **2.** | $ACBD$ is a paralellogram | If a quadrilateral has both pairs of opposite sides congruent, then it is a parallelogram. |
| **3.** | $\overline{CD}$ bisects $\overline{AB}$ | If a quadrilateral is a parallelogram, then its diagonals bisect one another. |
| **4.** | $\overline{AE} \cong \overline{BE}$ | Definition of a bisector |

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**d**

**To prove: $\overline{PR}$ is constructed so that $\angle TPR \cong \angle RPQ$**

| | Statements | Reasons |
|---|---|---|
| **1.** | $\overline{PT} \cong \overline{TR} \cong \overline{RS} \cong \overline{SP}$ | The arcs that created the segments have the same radius. |
| **2.** | $PTRS$ is a rhombus | Definition of a rhombus |
| **3.** | $\overline{RP}$ is a line bisector. | If a parallelogram is a rhombus, then each diagonal bisects a pair of opposite angles. |
| **4.** | $\angle TPR \cong \angle RPQ$ | Definition of an angle bisector |

**e**

**To prove: $\overline{PR}$ is constructed from line segment $\overline{PQ}$ such that $m\angle RPS = 30°$**

| | Statements | Reasons |
|---|---|---|
| **1.** | $\overline{PT} \cong \overline{TS} \cong \overline{SP}$ | The arcs that created the segments have the same radius. |
| **2.** | $\triangle PTS$ is a equilateral | Definition of an equilateral triangle |
| **3.** | $m\angle TPS = 60°$ | Each angle of an equilateral triangle measures $60^\circ$ |
| **4.** | $\overline{PT} \cong \overline{TR} \cong \overline{RS} \cong PS$ | The arcs that created the segments have the same radius. |
| **5.** | $PTRS$ is a rhombus | Definition of a rhombus |
| **6.** | $\overline{RP}$ is an angle bisector | If a parallelogram is a rhombus, then each diagonal bisects a pair of opposite angles. |
| **7.** | $\angle TPR \cong \angle RPS$ | Definition of an angle bisector |
| **8.** | $m\angle TPR = m\angle RPS$ | Definition of congruence |
| **9.** | $m\angle TPR + m\angle RPS = m\angle TPS$ | Angle addition postulate |
| **10.** | $m\angle RPS + m\angle RPS = m\angle TPS$ | Substitution |
| **11.** | $m\angle RPS + m\angle RPS = 60°$ | Substitution |
| **12.** | $2m\angle RPS = 60°$ | Simplify |
| **13.** | $m\angle RPS = 30°$ | Division property of equality |

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**2** A point $A$ is arbitrarily chosen along the circle. We create the diameter $\overline{AC}$ by passing it through the center of the circle at $O$. We choose a sufficiently large radius and construct the pair of arcs at $E$ and $F$ by centering them at $A$ and $C$. Then the line segments $\overline{AE}$, $\overline{CE}$, $\overline{CD}$ and $\overline{DA}$ are congruent since they have been constructed from the same radius. So $AECF$ is a rhombus.

We then construct $\overline{EF}$. The diagonals of a rhombus are perpendicular, so $\overline{EF}$ is perpendicular to $\overline{AC}$. $AECF$ is a rhombus, which means it's also a parallelogram. The diagonals of a parallelogram bisect one another, so $\overline{EF}$ bisects $\overline{AC}$. Since $\overline{AC}$ is a diameter of the circle, then $EF$ bisects $\overline{AC}$ through the center at $O$.

We then extend $\overline{EF}$ to the edges of the circle at $B$ and $D$. Then $\overline{BD}$ bisects and is perpendicular to $\overline{AC}$. $\overline{AC}$ also bisects $\overline{BD}$ since $\overline{BD}$ is a diameter as well. $\overline{BD}$ and $\overline{AC}$ are congruent since they are diameters of the same circle. So $ABCD$ is a square, since the diagonals are congruent, perpendicular and bisect one another. And finally, since the vertices lie on the circle, $ABCD$ is an inscribed square.

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